SOCIOL 723: Social Statistics II

Week 10, Panel Data and Difference-in-Differences

Wenhao Jiang
Department of Sociology, Fall 2026

Duke
UNIVERSITY

Today

Panel Data and Unobserved Heterogeneity

Fixed or Random Intercepts, Without Time

Difference-in-Differences

Staggered Adoption

Event Studies

TWFE Beyond a One-Time 0/1 Treatment

★ = essential, you must understand this ★ = for understanding, not examined

A Third Source of Variation

So far: adjust for observables (Week 6), find an instrument (Week 7), find a threshold (Week 9). Today: use variation over time within the same unit.

The core idea

Observe the same unit before and after treatment.

- What is **stable** about the unit (ability, upbringing) is the same in both periods, so the before–after change removes it, measured or not: **fixed effects**.
- What **changes for everyone** (a recession, getting older) does not cancel, but units that were not treated went through it too, so their change measures it: **difference-in-differences**.

One idea, one assumption: nothing else changes differently for the treated units.

Panel Data

The Example: Does Marriage Raise Men's Wages? ★

The question. Married men earn more than unmarried men. Is it marriage (specialization at home, employers' favour), or do men with high earning potential marry?

The data. NLSY men observed every year 1980–87: 545 men, 4,360 person-years (Vella and Verbeek 1998; `wooldridge::wagepan`). Y_{it}: log hourly wage of man i in year t; $D_{it} = 1$ if married that year.

men never married in the panel	162
men married in all eight years	73
men whose status changes	310
share married, 1980 → 1987	0.19 → 0.61
mean log wage, 1980 → 1987	1.39 → 1.87

Start simple: 1980 and 1987 only; the 528 men who married or did not change (17 who left marriage set aside). All men and years come later (figs/w10_panel_slow.R).

Pooled OLS ★

Treat the 1,056 person-years (528 men, 1980 and 1987) as if they were 1,056 different men:

$$Y_{it} = a + \beta D_{it} + u_{it}, \qquad \hat{\beta} = 0.358 \ (0.031)$$ married men earn about 43% more

$$\operatorname{plim} \hat{\beta} = \beta + \frac{\operatorname{Cov}(D_{it}, u_{it})}{\operatorname{Var}(D_{it})}$$ omitted variables (Week 2)

- u_{it} holds everything else that sets man i's wage in year t:
  - *stable* traits: ability, ambition, schooling, family background;
  - what *changes over time*: experience, seniority, prices, the economy.
- Both can move with marriage: men with high-wage traits may marry more, and men marry as they age while every man's wage grows. Either way $\operatorname{Cov}(D_{it}, u_{it}) > 0$ and 0.358 is too big.
- We cannot measure all of them. But we see each man more than once.

Standard errors clustered by man throughout (figs/w10_panel_slow.R).

Compare Each Man With Himself ★

Split the error in three: $u_{it} = \alpha_i + \lambda_t + \epsilon_{it}$.

- α_i: the combined effect on his wage of *everything stable* about man i (ability, ambition, schooling, family background), measured or not.
- λ_t: what changes for *every* man in year t (experience, prices).
- ϵ_{it}: the rest.

Take 1987 and 1980:

$$Y_{i,87} = \alpha_i + \lambda_{87} + \beta D_{i,87} + \epsilon_{i,87} \qquad \text{1987}$$

$$Y_{i,80} = \alpha_i + \lambda_{80} + \beta D_{i,80} + \epsilon_{i,80} \qquad \text{1980}$$

$$Y_{i,87} - Y_{i,80} = \underbrace{(\lambda_{87} - \lambda_{80})}_{\text{intercept}} + \beta\,(D_{i,87} - D_{i,80}) + (\epsilon_{i,87} - \epsilon_{i,80}) \qquad \text{subtract: } \alpha_i \text{ cancels}$$

First differences: regress each man's wage growth on his change in marital status, with an intercept. Idea 1: compare each man with himself. Every stable trait is gone, measured or not.

Men Who Did Not Change Are the Control Group ★

change in married, 1980 → 1987	men	mean wage growth
married ($\Delta D = 1$)	251	0.488
no change ($\Delta D = 0$)	277	0.476

OLS: $\hat{\lambda}_{87} - \hat{\lambda}_{80} = 0.476$ (0.037), $\hat{\beta} = 0.012$ (0.053).

- The intercept is the year effect, $\lambda_{87} - \lambda_{80}$, estimated from the 277 men whose status did not change. Idea 2: the year fixed effect makes them the control group.
- $\hat{\beta} = 0.488 - 0.476$ is the *extra* growth of men who married: difference-in-differences. If the timing of marriage were random, this is the comparison an experiment would make.
- Pooled OLS said 0.358. Most of it was stable traits and common growth.

All Men, All Years: The Model ★

Now all 545 men (the 17 who left marriage too) and every year, $t = 1980, \ldots, 1987$:

$$Y_{it} = \underbrace{\alpha_i}_{\text{who he is}} + \underbrace{\lambda_t}_{\text{what year it is}} + \underbrace{\beta D_{it}}_{\text{marriage}} + \underbrace{\epsilon_{it}}_{\text{everything else}} .$$

- α_i: one number per man, for everything stable about him (ability, ambition, family background), measured or not: man fixed effects.
- λ_t: one number per year, for what happens to every man that year (the economy, a year more experience): year fixed effects.
- With both sets of fixed effects in the model, people call it two-way fixed effects (TWFE).

Man Fixed Effects: Each Man His Own Line

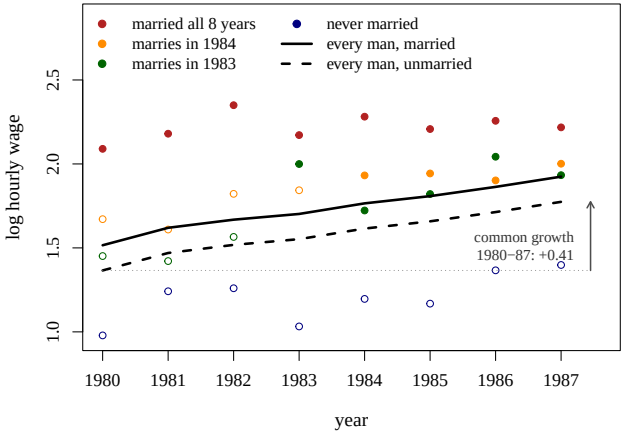
$Y_{it} = \alpha_i + \beta D_{it} + \epsilon_{it}$: one dummy per man, one common slope. Four of the men:



- The pooled line (slope 0.66 for these four) mostly compares *different* men: the high-wage men are the married ones.
- Each colored line is one man at his own height $\hat{\alpha}_i$. Its slope (0.31 for these four) compares him only with himself: married years vs. unmarried years.
- All 545 men: $\hat{\beta} = 0.243$ (0.022). But a man's married years are also his later years, when every wage is higher.

Year Fixed Effects: One Common Year Path

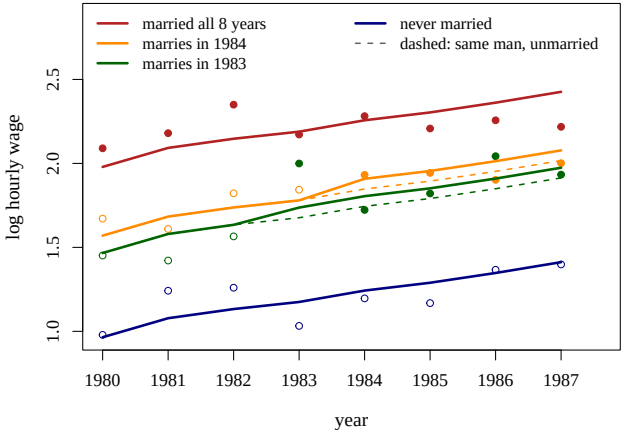
$Y_{it} = a + \lambda_t + \beta D_{it} + \epsilon_{it}$: one dummy per year, the same for every man.



- Dashed: $\hat{a} + \hat{\lambda}_t$, the wage of an unmarried man in year t. It rises 0.41 from 1980 to 1987: growth every man gets, married or not.
- Solid: the same path plus $\hat{\beta} = 0.150$ (0.028) in married years.
- But every man shares one line: the red man is high in every year, married or not, so $\hat{\beta}$ still compares different men.

Two-Way FE: Own Level, Common Year Path ★

$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it}$. Fitted on all 545 men; four of them shown:



- Every man follows the same year path, shifted to his own level.
- Marriage lifts him off his own path: the gap between solid and dashed.
- $\hat{\beta} = 0.061$ (0.022). Filled points: married years.

Two-Way FE Can Be Demeaned: the Algebra ★

Subtract the man's mean *and* the year's mean. In a balanced panel:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it} \qquad \text{the model}$$

$$\bar{Y}_{i\cdot} = \alpha_i + \bar{\lambda} + \beta \bar{D}_{i\cdot} + \bar{\epsilon}_{i\cdot} \qquad \text{average over } t\text{: man } i\text{'s mean}$$

$$\bar{Y}_{\cdot t} = \bar{\alpha} + \lambda_t + \beta \bar{D}_{\cdot t} + \bar{\epsilon}_{\cdot t} \qquad \text{average over } i\text{: year } t\text{'s mean}$$

$$\bar{Y} = \bar{\alpha} + \bar{\lambda} + \beta \bar{D} + \bar{\epsilon} \qquad \text{average over both}$$

$$Y_{it} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot t} + \bar{Y} = \beta(D_{it} - \bar{D}_{i\cdot} - \bar{D}_{\cdot t} + \bar{D}) + \ddot{\epsilon}_{it} \qquad \text{line } 1 - 2 - 3 + 4\text{: } \alpha_i \text{ and } \lambda_t \text{ cancel}$$

- **Double demeaning**: subtract the man's mean and the year's mean, add back the grand mean (subtracted twice). Same $\hat{\beta}$ as man and year dummies: 0.0606 both ways.
- With only 1980 and 1987 (528 men) it gives 0.012, the same as first differences.

How Software Does It: Dummies, or Means

Man 1 marries in 1982; man 2 never does. Omitted dummies: man 2 and 1980.

The data

man	year	Y	D	man 1	1981	1982
1	1980	1.5	0	1	0	0
1	1981	1.8	0	1	1	0
1	1982	2.4	1	1	0	1
2	1980	1.2	0	0	0	0
2	1981	1.5	0	0	1	0
2	1982	1.8	0	0	0	1

The man's mean of each column

Y	D	man 1	1981	1982
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$

- Subtract right from left, row by row: the man 1 column becomes all 0, so subtracting means does the man dummies' job.
- What is left of Y is each man's distance from his own average: TWFE compares each man with himself over time.
- Man 2 gives the year path (+0.3, +0.6); man 1 grew 0.9: $\hat{\beta} = 0.3$.

What Fixed Effects Averages: One Man's Own Estimate

Man effects only, binary D. Run the within regression for *one* man:

$$b_i = \text{(his mean wage in married years)} - \text{(his mean wage in unmarried years)},$$

because a regression on one 0/1 variable with an intercept is a difference in means (Week 1). **Example** (hypothetical): married in 3 of 8 years.

	log wages	mean
5 unmarried years	1.50, 1.55, 1.60, 1.62, 1.63	1.58
3 married years	1.70, 1.75, 1.80	1.75
b_i		$1.75 - 1.58 = 0.17$

- A man who never changes status has no b_i: he adds nothing.
- Fixed effects combines the 310 men's b_i into one number. The question is how it weights them.

The Weights, Step 1: Fixed Effects Is an Average of the b_i

Tilde: minus the man's own mean. Write man i's piece of the fixed-effects formula as $N_i = \sum_t \tilde{D}_{it}\tilde{Y}_{it}$ (numerator) and $S_i = \sum_t \tilde{D}_{it}^2$ (denominator):

$$\hat{\beta}_{FE} = \frac{\sum_i \sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} = \frac{N_1 + N_2 + \ldots}{S_1 + S_2 + \ldots} \qquad \text{add up man by man}$$

$$b_i = N_i/S_i, \quad \text{so } N_i = S_i\, b_i \qquad \text{man } i\text{'s own regression}$$

$$\hat{\beta}_{FE} = \frac{S_1 b_1 + S_2 b_2 + \ldots}{S_1 + S_2 + \ldots} = \sum_i w_i\, b_i, \qquad w_i = \frac{S_i}{S_1 + S_2 + \ldots} \qquad \text{substitute}$$

- Fixed effects is a weighted average of the men's own estimates b_i.
- Man i's weight is his S_i, the variation in his own marriage status, as a share of everyone's.

The Weights, Step 2: What a Man's S_i Is

Man i is married in m_i of T years, so $\bar{D}_i = m_i/T = p_i$. His demeaned 0/1 values:

$$\tilde{D}_{it} = 1 - p_i \qquad \text{in his } m_i \text{ married years}$$

$$\tilde{D}_{it} = -p_i \qquad \text{in his } T - m_i \text{ unmarried years}$$

$$S_i = m_i(1-p_i)^2 + (T-m_i)\,p_i^2 \qquad \text{square and add}$$

$$= Tp_i(1-p_i)^2 + T(1-p_i)\,p_i^2 = T\,p_i(1-p_i) \qquad m_i = Tp_i\text{; factor out } Tp_i(1-p_i)$$

Check: $T = 8$, married 2 years, $p = 0.25$:
$2(0.75)^2 + 6(0.25)^2 = 1.125 + 0.375 = 1.5 = 8(0.25)(0.75)$.

years married, of 8	1	2	3	4	5	6	7
weight per man, $8p(1-p)$	0.88	1.50	1.88	2.00	1.88	1.50	0.88
share of total weight	.08	.11	.19	.15	.22	.14	.12

Largest for men married half the time; a man married 1 of 8 years counts less than half as much.

The Weights, Step 3: Two Men

	married	own estimate b_i	weight $S_i = 8p(1-p)$
man A	4 of 8 years	0.10	2.000
man B	1 of 8 years	0.40	0.875

$$\hat{\beta}_{FE} = \frac{2.000 \times 0.10 + 0.875 \times 0.40}{2.000 + 0.875} = \frac{0.55}{2.875} = 0.19$$

$$\text{plain average} = \frac{0.10 + 0.40}{2} = 0.25$$

- Fixed effects leans toward man A, whose status varies more, not toward the typical man. If effects differ across men, FE is not the average effect.

man effects only: $\hat{\beta}_{FE} = 0.243$ = weighted mean of 310 men's b_i (unweighted 0.255). Adding year effects: 0.061; wages rise with age for everyone, and marriage comes with age (figs/w10_panel.R). Toy numbers (figs/w10_toy_staggered.R).

In the Literature: Budig and England (2001)

The question. Why do mothers earn less? Does motherhood lower wages, or do women with lower earning potential have more children?

The data. NLSY79 women 1982–93: 5,287 women, 41,842 person-years.

Their model (p. 213), "effects are fixed for years and persons":

$$\ln w_{it} = \beta \, \text{children}_{it} + X'_{it}\gamma + u_i + v_t + e_{it}$$

- children_{it}: number of children (0, 1, 2, ...), not a 0/1 treatment.
- u_i: person effects (aptitude, preferences, early socialization); v_t: year effects.
- X_{it} added in blocks: marital status; human capital (education, seniority, experience, breaks in employment); job characteristics.
- No age: person and year effects already fix it (age = year − birth year).

Budig and England: Results

Each row adds a block of controls; person and year effects in every fixed-effects model.

penalty per child, their Table 2	fixed effects	pooled OLS
person and year effects only	−0.068 (0.004)	−0.081 (0.002)
+ marital status	−0.073	−0.081
+ experience, seniority, education	−0.047	−0.018
+ job characteristics	−0.037	−0.012

- About 7% per child within women, close to OLS: little selection of lower-wage women into motherhood on fixed traits.
- Experience (lost years at work) explains about a third of the FE penalty; the rest remains. Note that experience is *post*-birth, a mediator: controlling for it answers a different question.

What Two-Way FE Assumes ★

The model, with the textbook's three assumptions (Wooldridge 2010, ch. 10):

$$Y_{it} = \beta D_{it} + \underbrace{\alpha_i}_{\text{stable: the man}} + \underbrace{\lambda_t}_{\text{common: the year}} + \underbrace{\epsilon_{it}}_{\text{the rest}} .$$

	assumption	needed for	marriage data
FE.1	strict exogeneity: $E[\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i] = 0$, all t	unbiased, consistent	tested below
FE.2	D_{it} varies over time within some units	identification	310 of 545 men
FE.3	$\text{Var}(\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i) = \sigma^2$ and $\text{Cov}(\epsilon_{it}, \epsilon_{is} \mid D_{i1}, \ldots, D_{iT}, \alpha_i) = 0$, $t \neq s$	usual s.e.	false: cluster by man

- Nothing is assumed about α_i: it may correlate with D_{it} in any way.
- Random effects adds $\text{Cov}(\alpha_i, D_{it}) = 0$, all t: men who marry, early or late, have the same stable traits on average as men who do not.

Strict Exogeneity, in Words ★

Two-way FE compares each man's married years with his *own average over all years*. So FE.1 asks: is a man's marriage in any year unrelated to his wage shocks in *every* year, past, present, and future?

- **Feedback** breaks it. A raise in 1983 ($\epsilon_{i,83}$ high) makes a man more likely to marry in 1984. The raise sits in his *unmarried* years and lifts his own baseline, so FE understates the effect; if men marry after a bad year, FE overstates it.
- **Something else changing at the same time** breaks it. A man who moves for a better job and marries that year: FE credits marriage with the new job.
- What it does *not* require: anything about α_i. Stable traits may be related to marriage in any way.

Feedback leaves a trace we can test: future marriage predicting today's wage.

Strict Exogeneity, Step by Step: The Estimator's Error ★

Man effects only, one regressor, T years. Tilde: minus the man's own mean.

$$
\begin{aligned}
Y_{it} &= \alpha_i + \beta D_{it} + \epsilon_{it} && \text{the model} \\
\bar{Y}_i &= \alpha_i + \beta \bar{D}_i + \bar{\epsilon}_i && \text{average over man } i\text{'s } T \text{ years} \\
\tilde{Y}_{it} &= \beta \tilde{D}_{it} + \tilde{\epsilon}_{it} && \text{line 1} - \text{line 2: } \alpha_i \text{ cancels} \\
\hat{\beta}_{FE} &= \frac{\sum_i \sum_t \tilde{D}_{it} \tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} && \text{OLS on the demeaned data} \\
&= \beta + \frac{\sum_i \sum_t \tilde{D}_{it} \tilde{\epsilon}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} && \text{substitute line 3 for } \tilde{Y}_{it}
\end{aligned}
$$

- Unbiased if the last fraction averages to zero: $\tilde{D}_{it}$ unrelated to $\tilde{\epsilon}_{it}$.
- But $\tilde{\epsilon}_{it} = \epsilon_{it} - \bar{\epsilon}_i$, and $\bar{\epsilon}_i = (\epsilon_{i1} + \cdots + \epsilon_{iT})/T$ holds *every* year's shock. So marriage in one year meets shocks from all years.

Testing Strict Exogeneity: Add the Lead

Under FE.1, next year's treatment cannot predict this year's outcome once α_i and this year's treatment are held fixed:

$$Y_{it} = \beta D_{it} + \delta D_{i,t+1} + X_{it}'\gamma + \alpha_i + \lambda_t + \epsilon_{it}, \qquad H_0 : \delta = 0.$$

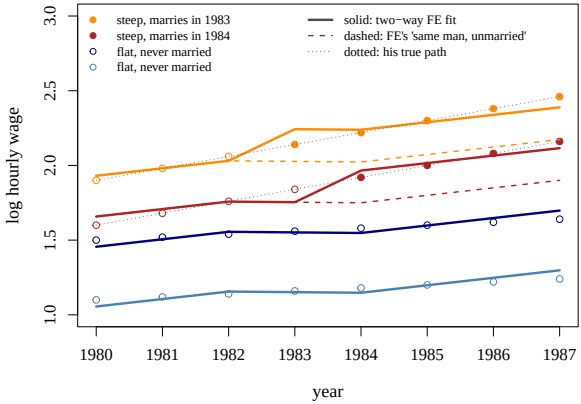
marriage data, 1980–86 (lead needs 1987)	estimate	(se)
married this year	0.048	(0.025)
married next year	−0.0005	(0.022)

- No sign that men marry after a wage jump: $p = 0.98$.
- The s.e. (0.022) is as large as the effect: a small feedback would go unseen.
- This is the first pre-period coefficient of an event study (Event Studies section).

Man and year effects, union, experience^2; clustered by man (figs/w10_panel_std.R).

When Men on Steeper Paths Marry

Hypothetical, no noise, and marriage has no effect at all. Fitted by two-way FE:



- Steep wage paths (0.08 a year) marry; flat paths (0.02) never do.
- α_i: each man his own *level*. λ_t: one *common* year path. No man gets his own growth.
- The common path is too flat for the steep men: after marriage they run above it, and FE calls the gap marriage, $\hat{\beta} = 0.22$.
- Parallel trends fails. Own slopes: $\hat{\beta} = 0$ (figs/w10_slopes.R).

In the Literature: Ludwig and Brüderl (2018)

The question. Is the male marriage wage premium causal, or do men on steeper careers marry? A test of specialization theory.

The data. NLSY79, 1979–2012: 4,287 employed men, 49,801 person-years; men observed never married at least once.

The design. Four estimators of one coefficient:

model	allows men to differ in	marriage premium
pooled OLS	nothing	17.7%
FE	wage level	8.3%
FE, group slopes	level; growth by future marital status	4.3%
FEIS	level and own wage growth	0.5% (n.s.)

Their Figure 3. Before marrying, future husbands' wages grew 5% per year of experience; never-married men's, 3%.

Random Trends: Each Man His Own Slope

FE allows men to differ in wage *levels*. If men on steeper wage paths marry, FE still mistakes their growth for a marriage effect. Give each man his own line:

$$Y_{it} = \alpha_i + \gamma_i\, x_{it} + \lambda_t + \beta D_{it} + \epsilon_{it} \qquad \text{own intercept } \alpha_i\text{, own slope } \gamma_i \text{ in experience } x$$

Year effects stay: own lines absorb differences in *growth* between men; λ_t absorbs shocks common to a year (a recession), which lie on no one's line.

Estimate β in three steps:

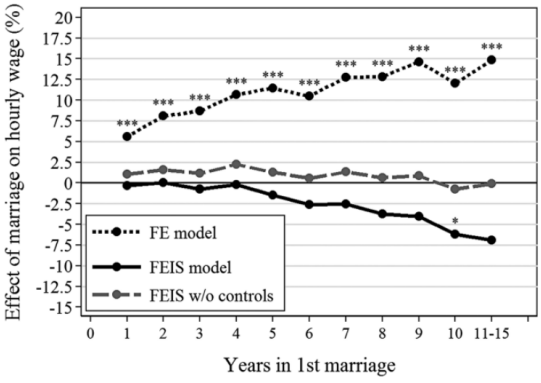
1. For each man, regress his wages on $(1, x_{it})$ using only his own years. Keep the residuals $\ddot{Y}_{it}$: his departures from his own line.
2. Do the same for marriage, giving $\ddot{D}_{it}$, and for each year dummy.
3. Pool all men and regress $\ddot{Y}_{it}$ on $\ddot{D}_{it}$ and the detrended year dummies.

Fixed effects is the same recipe with a line of slope zero: step 1 subtracts only his mean.

Random Trends: What Changes

- The **random trend model** (Wooldridge 2010, ch. 11); sociology calls it **FEIS**, fixed effects with individual slopes: detrend each unit separately, then pool (Brüderl and Ludwig 2015; Rüttenauer and Ludwig 2023). Needs at least three observations per unit.
- Identifies β from *departures* from each man's own trajectory at marriage: a jump, not a steeper path.
- Our 8-year panel: FE 0.047 (0.021), FEIS 0.055 (0.024). Eight years per man is short for estimating a slope; the NLSY79 over 34 years (next frames) tells a different story.

Ludwig and Brüderl: the Premium Over Time



Their Figure 4: effect by years since first marriage.

- FE: the premium grows to 11.4% after 5 years and 14.8% after 11–15, the pattern read as evidence of specialization.
- FEIS: about zero throughout.
- The growing "premium" is the steeper wage path the men were already on: selection on growth, not an effect of marriage.
- Generalizes: an FE impact function that keeps rising after the event is a warning sign of selection on slopes.

Two-Way Fixed Effects: Strengths and Weaknesses ★

Strengths

- Removes every stable trait of the unit, *measured or not*.
- Removes everything common to a year.
- Simple and transparent: a difference-in-differences with many units and periods.
- Testable in part: leads (no feedback), event studies.

Weaknesses

- Needs strict exogeneity: no feedback, nothing else changing at the same time.
- Misses differences in *growth*: men on steeper paths (FEIS).
- Only units whose treatment changes identify β, weighted by how much it varies.
- Less precise; measurement error grows. Cluster by unit.
- With a lagged outcome: Nickell bias (below).

Why Put Last Year's Outcome in a Fixed-Effects Model?

Marriage can follow wages, not only the other way round: a good year makes a man more likely to marry next year, and wages persist. So control for last year's wage:

$$Y_{it} = \rho\, Y_{i,t-1} + \beta D_{it} + \alpha_i + \epsilon_{it}$$

This compares men who start the year at the same wage, one who marries and one who does not; α_i still removes everything stable about the man.

Other questions where the outcome feeds back into the treatment:

- **Health and employment**: illness leads to job loss; job loss worsens health.
- **Parental help and test scores**: parents step in after a bad report card.
- **Policing and crime**: cities hire police after a crime spike; crime persists.

The catch: with α_i in the model, a lagged outcome breaks strict exogeneity.

Fixed Effects or a Lagged Outcome?

Sociologists often control for last year's outcome, to absorb differences between units that are *changing* rather than fixed. Combining it with unit fixed effects creates a new bias (Nickell, next frame).

- **FE and LDV embody different selection stories**: FE assumes selection on time-invariant traits; LDV assumes selection on prior outcomes. They can both be valid, but they are different assumptions and will generally give different answers.
- **Bracketing** (Angrist and Pischke 2009, sec. 5.4): for a positive effect, if LDV is the right model, FE overstates it; if FE is the right model, LDV understates it. Run both: the truth lies between.

Dynamic Panels: Nickell Bias

The outcome depends on its own past:

$$Y_{it} = \rho Y_{i,t-1} + X_{it}'\beta + \alpha_i + \epsilon_{it}.$$

FE.1 fails by construction: $Y_{i,t-1}$ contains $\epsilon_{i,t-1}$, which is in $\bar{\epsilon}_{i\cdot}$. Demeaned lag and demeaned error are correlated (Nickell 1981):

$$\text{plim}_{N\to\infty}\ \hat{\rho}_{FE} - \rho \ \approx\ -\frac{1+\rho}{T-1}$$

large-T approximation; exact formula in the script

simulation, $\rho = 0.5$, $N = 500$, 200 draws	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{FE}$	-0.04	0.27	0.40
Nickell's exact fixed-T value	-0.04	0.27	0.39

Large N does not help; only large T does (figs/w10_panel_std.R).

Anderson–Hsiao: Difference, Then Instrument

First-difference to remove α_i:

$$\Delta Y_{it} = \rho\,\Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta\epsilon_{it} \qquad \alpha_i \text{ gone}$$

$$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}, \quad \Delta\epsilon_{it} = \epsilon_{it} - \epsilon_{i,t-1} \qquad \text{both contain } \epsilon_{i,t-1}\text{: endogenous}$$

Instrument $\Delta Y_{i,t-1}$ with $Y_{i,t-2}$ (Anderson and Hsiao 1981), the Week 7 conditions:

- **Relevance**: $Y_{i,t-2}$ is part of $\Delta Y_{i,t-1}$.
- **Exogeneity**: $Y_{i,t-2}$ depends on errors up to $t-2$; $\Delta\epsilon_{it}$ on t and $t-1$. Holds only if ϵ_{it} is serially uncorrelated.

same simulation, $\rho = 0.5$	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{AH}$ (sd)	0.51 (0.13)	0.51 (0.05)	0.50 (0.03)

Arellano–Bond: Use Every Valid Lag

If ϵ_{it} is serially uncorrelated, *every* lag from $t-2$ back is a valid instrument for period t's differenced equation:

$$E\big[Y_{i,t-s}\,\Delta\epsilon_{it}\big] = 0, \qquad s \geq 2,\ t = 3, \ldots, T.$$

GMM combines all these moment conditions (Arellano and Bond 1991).

- More instruments, more precision: period t has $t-2$ of them. With $T = 8$ and year effects: 29 instruments.
- **The key check**: the differenced residuals $\Delta\hat{\epsilon}_{it}$. First-order correlation is expected: $\Delta\epsilon_{it}$ and $\Delta\epsilon_{i,t-1}$ share $\epsilon_{i,t-1}$. Second-order correlation must be zero; if it is not, ϵ_{it} is serially correlated and $Y_{i,t-2}$ is invalid. Start at $Y_{i,t-3}$ and check *third*-order correlation instead.
- Too many instruments overfit the first stage and pull GMM toward FE-style bias; report the count (Roodman 2009).

A Dynamic Wage Equation on the Marriage Data

log wage on its lag, married, union (both treated as strictly exogenous); year effects; robust s.e.; `fixest` (AH), `plm::pgmm` one-step (AB) (figs/w10_panel_std.R):

estimator	lagged wage $\hat{\rho}$	married	note
pooled OLS	0.60 (0.03)	0.056 (0.015)	α_i in the error: $\hat{\rho}$ too high
FE	0.06 (0.02)	0.062 (0.021)	Nickell: $\hat{\rho}$ too low
Anderson–Hsiao, Y_{t-2}	0.12 (0.03)	0.046 (0.022)	first-stage $F = 830$
Arellano–Bond, Y_{t-2} on	0.16 (0.04)	0.044 (0.023)	2nd-order test $z = 2.18$, $p = 0.03$
Arellano–Bond, Y_{t-3} on	0.68 (0.14)	0.031 (0.031)	3rd-order test $p = 0.97$
Anderson–Hsiao, Y_{t-3}	0.81 (0.27)	0.027 (0.034)	first-stage $F = 12$

- The second-order test rejects: wage shocks persist, so Y_{t-2} is not a valid instrument, and the rows using it are biased.
- Lag 3 passes the third-order test but is weak ($F = 12$): $\hat{\rho}$ of 0.7–0.8, above pooled OLS (an upper bound). Eight years do not pin down persistence.
- The marriage coefficient, 0.03–0.06, is far less sensitive than $\hat{\rho}$.

FE vs RE

Fixed or Random Intercepts: the Question

Goal: the same model whenever members share a group (students in a school, siblings in a family, workers in a firm, the years of one life):

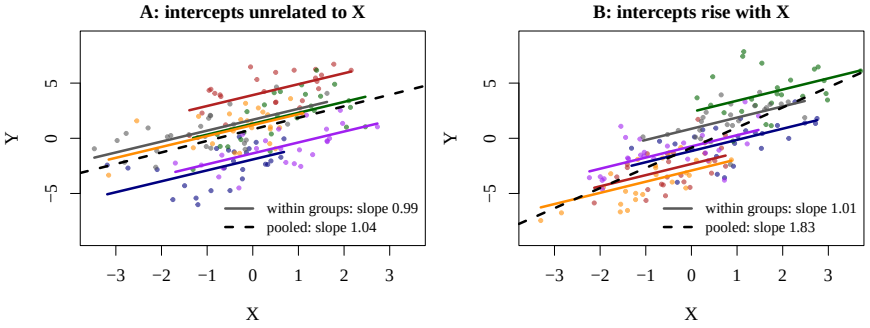
$$Y_{ij} = \alpha_j + \beta X_{ij} + \epsilon_{ij}.$$

	assumption on α_j	uses
fixed effects	none: α_j may covary with X	within groups only
random effects	$\text{Cov}(\alpha_j, X_{ij}) = 0$, all i: unrelated to X	within and between

- Both let α_j vary across groups. The names mislead: the key difference is whether α_j may covary with X (Wooldridge 2010, ch. 10).
- If it may not, random effects is more precise and estimates group-level variables.

Two Worlds in a Picture

Six groups, $\beta = 1$; same $\mathrm{Var}(\alpha_j)$ in both worlds, only its covariance with X differs (figs/w10_fe_sim.R):



- A: high-X groups are no higher; pooled and within slopes agree (1.04 vs 0.99).
- B: high-X groups have high intercepts; the pooled slope adds that gap (1.83 vs 1.01).

Three Estimators, One Formula

Start from the model, average it over the group's members, and subtract a share θ of that average:

$$Y_{ij} = \beta X_{ij} + \alpha_j + \epsilon_{ij} \qquad \text{the model}$$

$$\bar{Y}_j = \beta \bar{X}_j + \alpha_j + \bar{\epsilon}_j \qquad \text{average over group } j\text{'s members}$$

$$Y_{ij} - \theta \bar{Y}_j = \beta(X_{ij} - \theta \bar{X}_j) + (1-\theta)\,\alpha_j + (\epsilon_{ij} - \theta \bar{\epsilon}_j) \qquad \text{line 1 minus } \theta\times \text{ line 2}$$

θ	estimator	share of α_j left in the error
0	pooled OLS	all
between 0 and 1	random effects	$1-\theta$
1	fixed effects	none

- For $\theta < 1$ a share $(1-\theta)$ of α_j stays in the error. That is harmless only if α_j is unrelated to X: the random-effects assumption.

What θ Is: How Much to Trust the Group Mean

Random effects picks θ from two variances, $\tau^2 = \text{Var}(\alpha_j)$ and $\sigma^2 = \text{Var}(\epsilon_{ij})$, with n members per group:

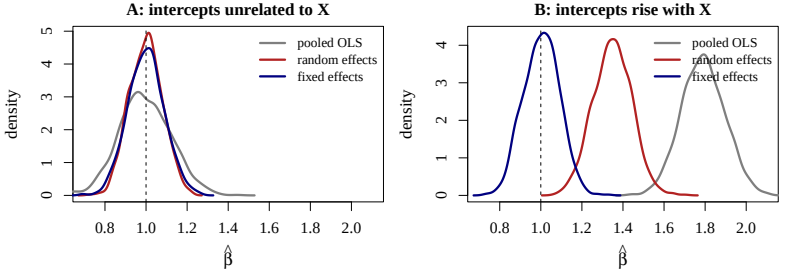
$$\theta = 1 - \sqrt{\frac{\sigma^2}{\sigma^2 + n\tau^2}} \qquad \text{(GLS; Wooldridge 2010, ch. 10).}$$

- The group mean $\bar{Y}_j$ carries α_j plus noise of variance σ^2/n.
- Many members or large group differences ($n\tau^2 \gg \sigma^2$): the group mean is mostly α_j, so remove nearly all of it, $\theta \to 1$, like fixed effects.
- Few members or small group differences: the group mean is mostly noise, so remove little of it, $\theta \to 0$, like pooled OLS.

$\sigma = 1.5$, $\tau = 2$	$n = 1$	$n = 4$ (simulation)	$n = 20$
θ	0.40	$1 - \sqrt{2.25/18.25} = 0.65$	0.84

1,000 Samples in Each World

100 groups of 4, $\beta = 1$:



mean (sd) of $\hat{\beta}$	pooled OLS	random effects	fixed effects
A: unrelated	0.995 (0.126)	0.999 (0.080)	1.000 (0.088)
B: rise with X	1.790 (0.105)	1.348 (0.094)	1.003 (0.087)

A: all unbiased, random effects most precise. B: random effects keeps $1 - \theta = 35\%$ of α_j and is biased; fixed effects is not.

Testing Fixed Against Random: Hausman

H_0: α_j is unrelated to X (Hausman 1978). Fixed effects is consistent either way; random effects is consistent and more precise under H_0, inconsistent otherwise.

$$H = \frac{(\hat{\beta}_{FE} - \hat{\beta}_{RE})^2}{\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})} \xrightarrow{d} \chi^2_1$$

- **Numerator**: how far apart the two estimates are. Under H_0 both are right, so they should differ only by noise.
- **Denominator**: how much noise to expect. Under H_0 random effects is the most precise, so $\text{Var}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) = \text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})$.
- Large H: the gap is more than noise, so reject random effects. In the 1,000 simulated samples it rejects 5.6% of the time in world A, 100% in world B.

Marriage data: $\chi^2_{10} = 37.0$, $p < 0.001$. Caveat: the denominator assumes random effects is fully efficient, which fails when errors are serially correlated.

On the Marriage Data: Random Effects

RE is OLS after subtracting a share θ of each man's mean. In a balanced panel, one θ for everyone:

$$Y_{it} - \theta \bar{Y}_{i\cdot} = (X_{it} - \theta \bar{X}_{i\cdot})'\beta + \ldots, \qquad \theta = 1 - \sqrt{\frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}}$$

$$\hat{\theta} = 1 - \sqrt{\frac{0.123}{0.123 + 8(0.105)}} = 0.64 \qquad \text{marriage data}$$

- The man effect is 46% of the error variance; RE removes 64% of each man's mean.
- RE needs FE.1 *and* $\mathrm{Cov}(\alpha_i, X_{it}) = 0$, all t: the man effect unrelated to his whole marriage history.
- RE's 0.064 lies between pooled OLS with controls (0.108) and FE (0.047): it partly uses the between-man comparison, and so can estimate education and race.

Swamy–Arora variance components, `plm` (figs/w10_panel_std.R).

Fixed or Random: How to Choose

if the question is	use
the effect of a member-level variable, and the group effect may be confounded	fixed effects
the effect of a group-level variable (school sector)	random effects (fixed effects cannot estimate it), which needs α_j unrelated to every regressor
how much variation lies between groups (Week 4's ICC)	random effects

- Not rejecting does not make fixed effects wrong: it stays consistent either way.
- Fixed effects is not automatically causal: it removes only what group members share.

DID

The Canonical 2×2 ★

Two groups, two periods. One group is treated in period 2.

	Before	After	Difference
Treated	$E[Y \mid g{=}1, t{=}1]$	$E[Y \mid g{=}1, t{=}2]$	Δ_1
Control	$E[Y \mid g{=}0, t{=}1]$	$E[Y \mid g{=}0, t{=}2]$	Δ_0
			$\tau = \Delta_1 - \Delta_0$

- The first difference removes anything fixed about each group.
- The second difference removes anything common to both groups over time.
- Equivalent regression: $Y_{it} = \alpha + \gamma\,\text{treated}_i + \delta\,\text{post}_t + \tau\,(\text{treated}_i \times \text{post}_t) + \epsilon_{it}$.
- Example from the panel section: wage growth 1980–87 was 0.488 for men who married and 0.476 for men who did not change, so $\tau = 0.012$.

The 2×2 in the Literature: Card and Krueger (1994)

New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992; Pennsylvania did not. 410 fast-food restaurants surveyed in February and November 1992.

FTE employment per store (se), their Table 3	PA (79)	NJ (331)	NJ − PA
February (before)	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
November (after)	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
change	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

- $2.76 = 0.59 - (-2.16)$: Pennsylvania's decline stands in for what New Jersey would have done without the law.
- Parallel trends here is one claim about one region in one season. Their within-NJ check: high-wage NJ stores, not bound by the law, changed like PA (−2.16 vs. −2.28, balanced sample).

The 2×2 in Regression Form

Group $g_i \in \{0, 1\}$, period $t \in \{1, 2\}$; the four cell means follow:

$$Y_{it} = \alpha + \gamma\, g_i + \delta\, \text{post}_t + \tau\, (g_i \times \text{post}_t) + \epsilon_{it}.$$

	Before	After	Difference
Treated ($g = 1$)	$\alpha + \gamma$	$\alpha + \gamma + \delta + \tau$	$\delta + \tau$
Control ($g = 0$)	α	$\alpha + \delta$	δ
			τ

- γ absorbs fixed differences in level; δ absorbs the common time trend; τ is what is left.
- Card and Krueger: $\alpha = 23.33$ (PA, February), $\gamma = -2.89$ (NJ minus PA in February), $\delta = -2.16$ (PA's change), $\tau = 2.76$.
- Unit and period fixed effects in place of g_i and post_t: two-way FE, identical with two groups and two periods, not otherwise.

Parallel Trends ★

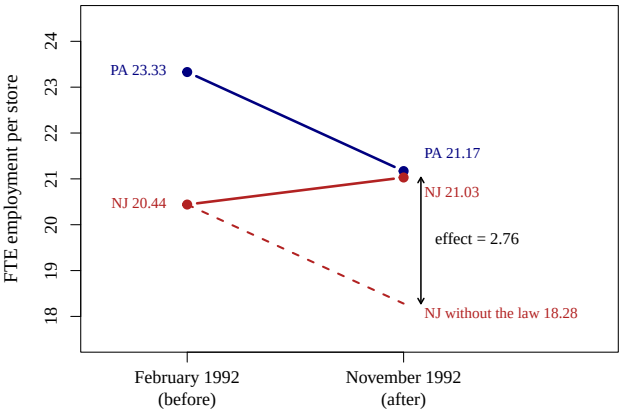
The identifying assumption

$$E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 0].$$

Absent treatment, the two groups' average outcomes would have moved in parallel.

- Levels may differ arbitrarily. Only the *trends* in the untreated potential outcome must agree.
- Untestable: it concerns a counterfactual trend. Pre-treatment parallelism is suggestive evidence, not proof.
- **Not scale-free**: parallel trends in levels and parallel trends in logs are different assumptions, and both cannot generally hold.
- Needs **no anticipation**: units must not change behaviour before treatment in expectation of it.

Parallel Trends in a Picture: Card and Krueger



- The assumption: without the law, NJ stores would have lost 2.16 workers, as PA stores did.
- Dashed: $20.44 - 2.16 = 18.28$, NJ without the law.
- Effect: $21.03 - 18.28 \approx 2.76$.
- Levels may differ (NJ started 2.89 lower); only the change must match.

Their Table 3; 2.75 from the rounded means (figs/w10_ck_parallel.R).

What the Double Difference Identifies, Exactly ★

Decompose the treated group's observed change. Its period-2 outcome is $Y_{i2}(1)$, its period-1 outcome is $Y_{i1}(0)$ (no anticipation); add and subtract the counterfactual $Y_{i2}(0)$:

$$\Delta_1 = E[Y_{i2}(1) - Y_{i1}(0) \mid g_i{=}1] = \underbrace{E[Y_{i2}(1) - Y_{i2}(0) \mid g_i{=}1]}_{\text{ATT}} + \underbrace{E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}1]}_{\text{treated group's } \textit{untreated} \text{ trend}}.$$

The control group supplies an estimate of that second, unobservable term: $\Delta_0 = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}0]$. Subtracting,

$$\boxed{\Delta_1 - \Delta_0 \;=\; \text{ATT} \;+\; \underbrace{E[\Delta Y(0) \mid g{=}1] - E[\Delta Y(0) \mid g{=}0]}_{\text{differential trend; } = 0 \text{ under parallel trends}}}$$

Reading the Double Difference ★

$\Delta_1 - \Delta_0 = \text{ATT} + \text{differential trend}$, so:

- The estimand is the ATT, not the ATE: we borrowed a counterfactual trend *for the treated group*. Nothing in the design identifies what treatment would have done to the controls.
- The bias term *is* the parallel-trends violation, exactly. Every diagnostic for DiD is aimed at that one object.
- Card and Krueger: NJ's change $0.59 = \text{ATT} +$ NJ's untreated trend. PA says that trend is -2.16, so ATT $= 2.76$. Had NJ's untreated trend really been -1.00, the DiD would still say 2.76 but the ATT would be $0.59 + 1.00 = 1.59$: the bias term is $-1.00 - (-2.16) = 1.16$.

Parallel Trends Is Not Scale-Free ★

Suppose untreated outcomes would have grown by 10% in both groups, from a base of 20 in the treated group and 40 in the control group.

	Before	After	Change
Treated	20	22	+2
Control	40	44	+4

- Parallel trends holds in logs (both +10%) but fails in levels (+2 versus +4). A DID in levels would report a spurious −2.
- Here both hold only if base levels are equal or nothing grows. There is no neutral choice: the transformation is part of the identifying assumption.
- Choose before looking at results: would untreated outcomes grow by the same amount, or the same percentage? It matters most when levels are far apart.

When Does the Scale Not Matter? (Roth and Sant'Anna 2023a)

Parallel trends in levels, in logs, in ranks: when can all hold at once?

Result

Parallel trends holds for *every* monotone transformation of Y if and only if the whole *distribution* of $Y(0)$ satisfies a parallel-trends condition. In practice: the population splits into a part where treatment is as good as random and a part whose $Y(0)$ distribution does not change over time.

- In any real DiD the scale is therefore part of the assumption. Choose it before seeing results, on substantive grounds (previous frame).
- The condition can be falsified in the pre-period: compare the groups' outcome distributions, not only means.
- For effects on whole distributions under weaker assumptions: Callaway and Li (2019).

Staggered

The Setting Everyone Uses ★

The 2×2 with many periods: units adopt the policy at *different times* (states pass a law in 1995, 2001, 2008). The standard specification is the 2×2 regression, extended:

$$Y_{it} = \alpha_i + \lambda_t + \beta\, D_{it} + \epsilon_{it}, \qquad D_{it} = \mathbb{1}\{t \geq G_i\}.$$

For thirty years this was assumed to estimate "the" treatment effect. It does when the effect is constant across units and over time since treatment. With heterogeneous or dynamic effects it need not estimate any sensible average.

Goodman-Bacon (2021)

The TWFE estimate is a weighted average of *all possible* 2×2 DID comparisons in the data, including comparisons that use already-treated units as controls.

The Marriage Data Are Mostly a Staggered Design ★

The panel section's TWFE (545 men, 1980–87, $\hat{\beta} = 0.061$) is this setting: men marry in different years. Each man falls into one group (figs/w10_marriage_timing.R):

group	men	role in TWFE
never married, 1980–87	162	never treated
married all 8 years	73	always treated
marry in 1981, …, 1987 and stay married	239	staggered: 51, 33, 40, 27, 30, 27, 31
marry, then leave marriage	71	switches off: not staggered

- Staggered adoption needs D to stay on once it starts. That holds for 474 of the 545 men; the 71 who leave marriage are an on/off design (Beyond 0/1 section).
- TWFE compares, for example, men who marry in 1985 with men who stay single (clean), and with men who married in 1981 (already treated).
- So the 0.061 rests on parallel trends *and* on a marriage premium that does not grow or fade with years married. If it does, the forbidden comparison bites.

Running Example: Unilateral Divorce and Female Suicide

The policy (Stevenson and Wolfers 2006). Before about 1970, most states granted a divorce only if both spouses agreed or one proved fault. *Unilateral* divorce lets either spouse end the marriage alone.

The question. Did the right to leave reduce women's suicide? A spouse who can leave has more say in the marriage; a woman in an abusive marriage gains a way out.

The data (`bacondecomp::divorce`): 50 states + DC, 1964–96. Y_{st} = female suicides per million women (51 in 1964); $D_{st} = 1$ from the year state s adopted:

adopted	states	role in the data
before 1964	9	always treated: no "before"
1969–72 (e.g. California 1970)	14	
1973–77	20	staggered: before and after observed
1980–85 (e.g. Pennsylvania 1980)	3	
not by 1996 (e.g. New York)	5	never treated

Their finding: female suicide fell about 10% on average over 20 years (event study, percent change); nothing for men (figs/w10_divorce.R).

The Forbidden Comparison: a Toy Panel ★

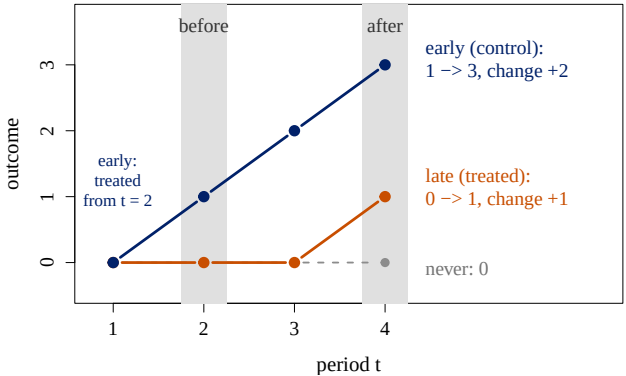
Before the real data, a toy with no noise. Three units, four periods. The outcome without treatment is 0 for everyone, so every number above 0 is a treatment effect:

unit	$t = 1$	$t = 2$	$t = 3$	$t = 4$	
early	0	**1**	**2**	**3**	treated from $t = 2$; effect grows 1, 2, 3
late	0	0	0	**1**	treated from $t = 4$; effect 1
never	0	0	0	0	never treated

- Bold cells are treated. Early's effect keeps growing the longer it is treated.
- Every effect is positive. Late's effect in $t = 4$ is exactly 1.

The Forbidden Comparison: Late vs. Early ★

A 2×2 that TWFE uses: **late** is the treated unit, **early** the control. Before $= t = 2$ (late untreated); after $= t = 4$ (late treated).



Early is treated at both dates, so its line rises for one reason: its own effect grows from 1 to 3.

(figs/w10_forbidden.R)

Late vs. Early, $t = 2 \to 4$: the Numbers ★

	before: $t = 2$	after: $t = 4$	change
late (treated)	0	1	+1
early (control)	1	3	+2
2×2 = late's change − early's change			$1 - 2 = -1$

- The control's change should be what late would do *untreated*: 0 (see never). Early rises by 2 only because *its own effect* grows from 1 to 3.
- Subtracting that growth gives −1, though every effect is positive. Late vs. never over the same dates gives the right answer: $1 - 0 = 1$.
- If early's effect were constant (1, 1, 1), its change would be 0 and the comparison harmless: the problem needs effects that change over time.

The Goodman-Bacon Decomposition ★

TWFE is a weighted average of every 2×2 in the data:

$$\hat{\beta}^{\text{TWFE}} = \sum_{\text{pairs}} s_{ab}\, \hat{\beta}^{2\times 2}_{ab}, \qquad \sum s_{ab} = 1.$$

- **The 2×2s**: treated vs. never (clean); earlier vs. later, before later switches (clean); later vs. earlier, after earlier switches (forbidden).
- **The weights**: how much data the pair uses $\times$ variance of D in the pair, the same rule as the panel section's $Tp_i(1-p_i)$.

Where a Weight Comes From: Early vs. Late

Early (treated) vs. late (control), periods 1–3, while late is still untreated:

	$t=1$	$t=2$	$t=3$
early: D	0	1	1
early: Y	0	1	2
late: D and Y	0	0	0

$$2\times 2 = \big[(1+2)/2 - 0\big] - \big[0 - 0\big] = 1.5 \qquad \text{before } t=1, \text{ after } t=2,3$$

$$\text{variance of } D = p(1-p) = \tfrac{2}{3}\cdot\tfrac{1}{3} = 0.222 \qquad \text{early has } D=1 \text{ in } p=2 \text{ of 3 periods}$$

$$\text{size} = (3/4)^2 = 0.5625 \qquad \text{uses 3 of the 4 periods, squared}$$

$$\text{raw weight} = 0.5625 \times 0.222 = 0.125$$

$p(1-p)$ is the panel section's $p_i(1-p_i)$: largest when the treated group is treated half the time. Equal group sizes, so they cancel (Goodman-Bacon 2021; figs/w10_toy_staggered.R).

The Decomposition on the Toy Example

Same steps for every pair; weight = raw weight / sum:

comparison	2×2	periods	$p(1-p)$	raw weight	weight
early vs. never	2.0	1–4	$\frac{3}{4} \cdot \frac{1}{4}$	$1 \times 0.1875 = 0.1875$	0.3
late vs. never	1.0	1–4	$\frac{1}{4} \cdot \frac{3}{4}$	$1 \times 0.1875 = 0.1875$	0.3
early vs. late	1.5	1–3	$\frac{2}{3} \cdot \frac{1}{3}$	$\frac{9}{16} \times 0.222 = 0.125$	0.2
late vs. early	-0.5	2–4	$\frac{1}{3} \cdot \frac{2}{3}$	$\frac{9}{16} \times 0.222 = 0.125$	0.2
sum				0.625	1

- Weighted sum: $0.3(2) + 0.3(1) + 0.2(1.5) + 0.2(-0.5) = 1.10$. The TWFE regression on the 12 cells gives 1.10; `bacon()` gives the same pieces.
- True average effect on the treated cells: 1.75 (figs/w10_toy_staggered.R).

The Goodman-Bacon Weights: What They Imply ★

- Every weight has the $p(1-p)$ shape of a Bernoulli variance, so groups treated near the middle of the panel dominate: the weights come from OLS mechanics, not from group sizes alone and not from any estimand you chose (Week 1's variance weighting).
- The forbidden comparison uses the periods after early switches. Its weight is larger when early switched early (a long stretch) and late switches halfway through it ($p = 1/2$). That stretch holds early's long-run effects: the ones it subtracts.
- All of this is computable from the data alone: `bacondecomp::bacon()` lists every 2×2, its estimate, and its weight.

The Decomposition on the Divorce Data

TWFE: $\hat{\beta} = -3.05$ (2.32). All 156 2×2 comparisons, grouped (figs/w10_divorce.R; Goodman-Bacon's own numbers, age-adjusted rates, 49 states, in grey):

comparison	weight	average 2×2	weight	average
timing states vs. always treated (pre-1964)	0.41	−6.92	0.38	−7.04
treated vs. never treated	0.23	−5.21	0.24	−5.33
earlier treated vs. later (not yet) treated	0.11	1.25	0.11	−0.19
later treated vs. earlier (already) treated	0.25	3.36	0.26	3.51
TWFE = weighted sum		−3.05		−3.08

- A quarter of the weight sits on comparisons that use already-treated states as controls; their average has the *opposite sign*.
- 41% comes from always-treated states as controls: their whole period is post-treatment, so they are clean only if their effects are constant.

Forbidden Comparisons Are the Negative Weights

de Chaisemartin and D'Haultfœuille (2020) write TWFE as a weighted average of the effects in the treated cells, and find that some weights are negative.

The forbidden comparison is where the negative weights come from

In the late-vs.-early 2×2, early is the control, so its $t = 4$ effect is *subtracted*. Collected cell by cell, that minus sign is a negative weight on early's $t = 4$ effect.

	diagnosis	solution
Goodman-Bacon (2021)	TWFE as weighted 2×2s; shows the forbidden ones (`bacon`)	use only clean 2×2s, e.g. Callaway–Sant'Anna
de Chaisemartin and D'Haultfœuille (2020)	TWFE as weighted cells; counts the negative weights (`twowayfeweights`)	compare units that switch at t only with units untreated at both $t-1$ and t (never or not yet treated; `DIDmultiplegt`)

The Cell Weights, Derived: TWFE as One Regression ★

TWFE is OLS: choose α_i, λ_t, β to minimize the squared errors,

$$\min_{\alpha,\lambda,\beta} \sum_{it} \left(Y_{it} - \alpha_i - \lambda_t - \beta D_{it}\right)^2.$$

The solution for β (balanced panel):

$$\tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}$$

$$\hat{\beta}^{\text{TWFE}} = \frac{\sum_{it} \tilde{D}_{it} Y_{it}}{\sum_{it} \tilde{D}_{it}^2}$$

Two facts about $\tilde{D}$:

- Its sum over t is 0 for every unit, and its sum over i is 0 for every period (demeaning in both directions).
- So anything that depends only on i, or only on t, has zero product with it: $\sum_{it} \tilde{D}_{it}\, a_i = \sum_i a_i \sum_t \tilde{D}_{it} = 0$.

The Cell Weights, Derived: Plug In the Outcome ★

Plug in $Y_{it} = \alpha_i + \lambda_t + \tau_{it} D_{it} + \epsilon_{it}$:

$$\sum \tilde{D}_{it} Y_{it} = \underbrace{\sum \tilde{D}_{it} \alpha_i}_{0} + \underbrace{\sum \tilde{D}_{it} \lambda_t}_{0} + \sum \tilde{D}_{it} \tau_{it} D_{it} + \underbrace{\sum \tilde{D}_{it} \epsilon_{it}}_{\to 0} \quad \text{previous slide}$$

$$\to \textstyle\sum_{D_{it}=1} \tilde{D}_{it}\, \tau_{it} \quad \text{only treated cells have } D = 1$$

$$\sum \tilde{D}_{it}^2 = \sum \tilde{D}_{it} D_{it} = \textstyle\sum_{D_{it}=1} \tilde{D}_{it} \quad D - \tilde{D} = \text{means: zero product}$$

Divide:

$$\text{plim } \hat{\beta}^{\text{TWFE}} = \sum_{D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{D_{it}=1} \tilde{D}_{it}}.$$

A treated cell's weight is negative exactly when its $\tilde{D}_{it} < 0$.

Negative Weights on the Toy Example ★

$\tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}$. Unit means: early 3/4, late 1/4; period means: 0, $\frac{1}{3}$, $\frac{1}{3}$, $\frac{2}{3}$; grand mean $\frac{1}{3}$.

treated cell	true effect τ_{it}	$\tilde{D}_{it}$	weight w_{it}	$w_{it}\tau_{it}$
early, $t = 2$	1	$1 - \frac{3}{4} - \frac{1}{3} + \frac{1}{3} = 0.25$	0.3	0.3
early, $t = 3$	2	0.25	0.3	0.6
early, $t = 4$	3	$1 - \frac{3}{4} - \frac{2}{3} + \frac{1}{3} = -0.083$	-0.1	-0.3
late, $t = 4$	1	$1 - \frac{1}{4} - \frac{2}{3} + \frac{1}{3} = 0.417$	0.5	0.5
sum		0.833	1	1.10

Same cell weights as the 2×2 expansion. Early's last period has the largest effect (3) and a negative weight: TWFE reports 1.10, the true average is 1.75 (figs/w10_toy_staggered.R).

Negative Weights: The Intuition ★

- **Which cells**: early adopters, late in the panel. In the toy, early at $t = 4$: treated in most periods, at a time when most units are treated.
- **Why**: TWFE learns from treatment that is *unexpected* given the unit and the period ($\tilde{D}$). Early at $t = 4$ is fully expected, so OLS does not use it as a treated cell. It uses it as a *control* for units switching later, and subtracts its effect.
- **When it hurts**: with a constant effect, adding and subtracting the same number cancels. When effects grow over time, the subtracted effect is the largest one (early at $t = 4$: 3), so TWFE is pulled down: 1.10, not 1.75.

Negative Weights on the Divorce Data

The same weight formula, computed cell by cell: $w_{it} = \tilde{D}_{it} / \sum_{D_{it}=1} \tilde{D}_{it}$ with $\tilde{D}$ the double-demeaned treatment (figs/w10_divorce.R).

treated state-years	1,164
with a negative weight	267
sum of the negative weights	−0.38
adopted before 1964 (always treated)	207 cells, −0.36
adopted 1969, 1970	60 cells, −0.02

- Early adopters, late in the panel (1974–96): their long-run effects, the largest in Stevenson and Wolfers, enter with the wrong sign.
- Always-treated weights sum to 0: they count only as controls, their late effects subtracted.
- Positive weights sum to 1.38, so TWFE is not a convex average of any group's effect.

Negative Weights: Consequences ★

de Chaisemartin and D'Haultfœuille (2020) generalize this calculation, allowing covariates and non-binary treatments, and show that negative weights are the rule in staggered designs, not a corner case.

- Consequence: TWFE can have the wrong sign even when every $\tau_{it} > 0$. This is not a small-sample problem; it survives $n \to \infty$, because the weights, not the noise, are at fault.
- By the formula, the negative weights fall on units treated *early* and observed over a *long* post-period, exactly the units carrying the most information about long-run effects.

TWFE is fine when treatment timing is common, or when effects are genuinely constant. It is unreliable exactly where sociologists use it most.

Negative Weights: Diagnostics ★

- **Diagnostic**: compute the weights (`TwoWayFEWeights::twowayfeweights`) and report how many are negative and what share of the total mass they carry. Small negative mass is reassuring about *this particular* problem; it says nothing about parallel trends or about dynamic-effect contamination, so it is one diagnostic, not a validity test.

The Fix: Clean 2×2s, Then One Average ★

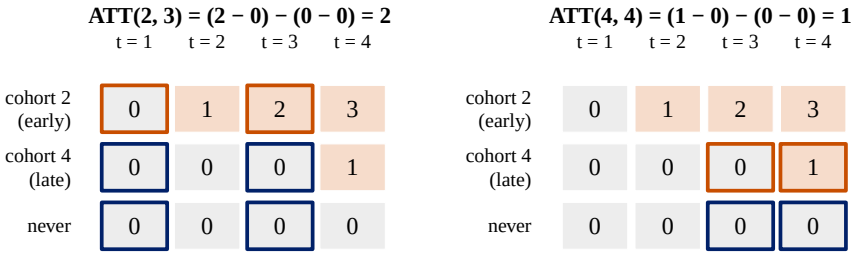
All of them share one principle: never use an already-treated unit as a control. Build clean 2×2 comparisons, then aggregate with weights you choose.

Method	Idea	R
Callaway–Sant'Anna (2021)	ATT(g, t); never- or not-yet-treated controls	`did`
Sun–Abraham (2021)	CS's blocks as one regression (Event Studies)	`fixest::sunab`
de Chaisemartin–D'Haultfœuille	clean comparisons at switch dates	`DIDmultiplegt`
Borusyak–Jaravel–Spiess	impute $Y(0)$ from untreated cells	`didimputation`

Callaway–Sant'Anna: Cohorts and Blocks

A **cohort** is all units that start treatment in the same period: the states that adopted in 1970 are cohort 1970. Toy: cohort 2 (early), cohort 4 (late), never treated.

Idea: one effect per cohort per period, $ATT(g, t)$, each from its own clean 2×2:



Orange: cohort g at $g-1$ and at t. Blue: units still untreated at t (controls). Shaded = treated (figs/w10_cs_blocks.R).

Callaway–Sant'Anna: Every Block on the Toy

$$ATT(g,t) = \underbrace{(Y_t - Y_{g-1})_{\text{cohort } g}}_{\text{its change}} - \underbrace{(Y_t - Y_{g-1})_{\text{not yet treated at } t}}_{\text{controls' change}}$$

block	cohort, periods	controls	$ATT(g,t)$
$ATT(2,2)$	early, $1 \to 2$	late, never	1
$ATT(2,3)$	early, $1 \to 3$	late, never	2
$ATT(2,4)$	early, $1 \to 4$	never (late is treated at 4)	3
$ATT(4,4)$	late, $3 \to 4$	never	1
average			1.75

- Every block recovers its true effect; the average is the true 1.75 (TWFE: 1.10).
- Needs parallel trends for each cohort with its controls (figs/w10_cs_blocks.R).

Callaway–Sant'Anna: Then Aggregate

The blocks $ATT(g,t)$ are the building pieces. **Choose how to average them**; each choice is a different question:

- *simple*: average over all treated (g,t), one number (toy: 1.75);
- *dynamic*: average by years since start, $t-g$: an event-study path (next section);
- *group*: average within each cohort: who benefited;
- *calendar*: average within each period: when.

```
did::aggte(cs, type = "simple" | "dynamic" | "group" | "calendar").
```

Covariates Inside Each Block

Parallel trends may hold only among similar units: South and non-South states trend differently. Each block is a 2×2, so adjust it like Week 6. Hypothetical block, true effect 1:

	South	non-South	mean change	
cohort g (3 South, 1 non-South)	+5	+3	4.5	
controls (1 South, 3 non-South)	+4	+2	2.5	unadjusted: 2.0

- **Outcome regression**: predict each cohort unit's untreated change from controls *like it* (South +4, non-South +2): $4.5 - \frac{3(4)+2}{4} = 4.5 - 3.5 = 1$.
- **IPW**: reweight controls to the cohort's mix (South $\times 3$, non-South $\times \frac{1}{3}$): $4.5 - 3.5 = 1$.
- **Doubly robust**: both; right if either model is right.

`att_gt(..., xformla = ~ south, est_method = "dr")` (or `"reg"`, `"ipw"`)

(figs/w10_cs_blocks.R).

Imputation, Step by Step (Borusyak, Jaravel, and Spiess 2024)

Use only untreated cells to learn the untreated model, then fill in:

1. $Y_{it} = \alpha_i + \lambda_t + \epsilon_{it}$ on cells with $D_{it} = 0$ — never-treated, and pre-periods of the treated
2. $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$ for cells with $D_{it} = 1$ — impute the missing potential outcome
3. $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0), \quad \hat{\tau} = \sum_{D_{it}=1} v_{it}\,\hat{\tau}_{it}$ — average with weights you choose

- No treated cell is ever a control; any aggregation (v_{it}) is allowed, so the estimand is chosen, not inherited from OLS.
- **Efficient** when errors are homoskedastic and serially uncorrelated: it uses all pre-periods, not just $g - 1$.
- **Pre-trend tests are separate**: run on untreated cells only, so testing does not change the estimate.
- Gardner (2022)'s two-stage DiD is the same idea as two regressions (`did2s`).

Imputation on the Toy Example

Untreated outcome 0 everywhere; bold cells are treated:

Y_{it}	$t=1$	$t=2$	$t=3$	$t=4$
never treated	0	0	0	0
early (from $t=2$; effects 1, 2, 3)	0	**1**	**2**	**3**
late (from $t=4$; effect 1)	0	0	0	**1**

1. Fit $\alpha_i + \lambda_t$ on the 8 untreated cells. All are 0, so every $\hat{\alpha}_i = \hat{\lambda}_t = 0$.
2. Impute $\hat{Y}_{it}(0) = 0$ in the 4 treated cells.
3. $\hat{\tau}_{it} = Y_{it} - 0$: 1, 2, 3 and 1. Average: $7/4 = 1.75$, the true average effect on the treated cells.

TWFE on the same 12 cells: 1.10; without the never-treated unit: 0.50 (figs/w10_toy_staggered.R).

The Estimators on One Dataset

Divorce data without the 9 always-treated states (they have no pre-period); standard errors clustered by state (figs/w10_divorce.R):

estimator	controls, baseline	ATT	(se)
TWFE, all 51 states	everything	−3.05	(2.32)
TWFE, 42 states	everything	−0.34	(2.76)
Callaway–Sant'Anna	never treated, year $g-1$	−10.56	(3.12)
Callaway–Sant'Anna	not yet treated, year $g-1$	−8.69	(3.40)
imputation	not yet treated, all pre-years	−4.85	(2.97)

- The robust estimators agree on the sign and say TWFE understates the effect.
- They differ in size because they assume parallel trends over different windows ($g-1$ only, or every pre-period) and weight the (g,t) cells differently.

Event Studies

Event Studies ★

D_{it} is still **binary**: 0 before unit i starts, 1 from then on. The static TWFE puts one β on D_{it}: every cohort and every year since the start, averaged into one number. The event study estimates one effect per period relative to the start:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - G_i = k\} + \epsilon_{it},$$

where G_i is unit i's treatment date (for example, the year a man marries) and $k = -1$ is the omitted reference.

- **Pre-period coefficients** ($k < 0$) are the parallel-trends check. Flat and near zero is what you want to see.
- **Post-period coefficients** trace out the dynamics: does the effect grow, fade, or appear only after a lag?
- Caution: with staggered G_i, the event study can inherit the forbidden comparison (end of this section).

Each τ_k Is a 2×2 Against Period $k = -1$ ★

Treated units start at $G = 0$, so $k = t$; controls never start. Plug in, dropping ϵ:

	$t = -2$	$t = -1$ (base)	$t = 0$ (start)	$t = 1$
treated	$\alpha_T + \lambda_{-2} + \tau_{-2}$	$\alpha_T + \lambda_{-1}$	$\alpha_T + \lambda_0 + \tau_0$	$\alpha_T + \lambda_1 + \tau_1$
control	$\alpha_C + \lambda_{-2}$	$\alpha_C + \lambda_{-1}$	$\alpha_C + \lambda_0$	$\alpha_C + \lambda_1$

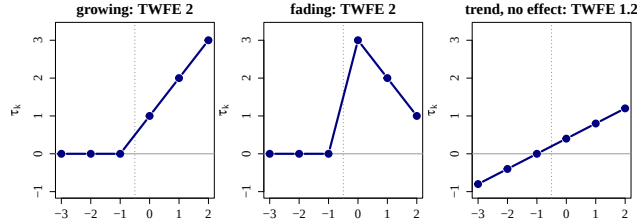
Two-period DiDs against $t = -1$ (α cancels in a row, λ across rows):

$$\begin{aligned}
\text{pre, } t = -2 \text{ vs. } -1: &\quad (\lambda_{-2} - \lambda_{-1} + \tau_{-2}) - (\lambda_{-2} - \lambda_{-1}) = \tau_{-2} \\
\text{start, } t = 0 \text{ vs. } -1: &\quad (\lambda_0 - \lambda_{-1} + \tau_0) - (\lambda_0 - \lambda_{-1}) = \tau_0 \\
\text{post, } t = 1 \text{ vs. } -1: &\quad (\lambda_1 - \lambda_{-1} + \tau_1) - (\lambda_1 - \lambda_{-1}) = \tau_1
\end{aligned}$$

- An event study is a set of two-period DiDs that share one base period, $t = -1$.
- **Pre**: τ_{-2} should be 0; if not, the groups were already drifting apart. **From** $t = 0$ **on**: the usual DiD effect; $t = 0$ is already treated.

The Event Study on the Toy Example ★

Toy: 2 treated, 2 control units, periods 1–6; D: $0 \to 1$ at period 4. Titles: the one-number TWFE $\hat{\beta}$ (figs/w10_twfe_many.R):



- **Growing vs fading**: same $\hat{\beta}$, opposite paths; the event study separates them.
- **Trend**: leads $-0.8, -0.4$ already slope up, and the "effects" $0.4, 0.8, 1.2$ continue the same line. The picture warns you; $\hat{\beta} = 1.2$ did not. This is Ludwig and Brüderl's marriage premium: future husbands were already on steeper paths; FE 8.3% and rising, FEIS 0.5%.
- More transparent, not more causal: the leads check only the pre-period; the lags are effects only if trends would have stayed parallel *after* adoption, which no data can show.

Reading the Event-Study Equation

One man married in 1984 ($G_i = 1984$), observed 1980–87. $k = t - G_i$ is years since marriage:

year t	1980	1981	1982	1983	1984	1985	1986	1987
$k = t - 1984$	-4	-3	-2	-1	0	1	2	3
dummy that is 1	τ_{-4}	τ_{-3}	τ_{-2}	none	τ_0	τ_1	τ_2	τ_3

- Each year switches on exactly one dummy, except $k = -1$, the reference: every τ_k compares year k with the year before marriage.
- A man who never marries has every dummy 0: he only helps estimate α_i and λ_t.
- $\tau_{-4}, \tau_{-3}, \tau_{-2}$: marriage cannot have caused anything yet, so they should be near 0.

In the Literature: Kleven, Landais, and Søgaard (2019)

The question. How much of the gender gap in earnings is caused by children? Danish registers 1980–2013; parents with a first birth 1985–2003, followed from 5 years before to 10 years after.

The design: an event study around the first birth, separately for mothers and fathers, with age and year dummies:

$$Y_{ist}^g = \sum_{j \neq -1} \alpha_j^g \mathbb{1}\{j = t\} + \sum_k \beta_k^g \mathbb{1}\{k = \text{age}_{is}\} + \sum_y \gamma_y^g \mathbb{1}\{y = s\} + \nu_{ist}^g$$

$$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w)/E[\tilde{Y}_{ist}^w \mid t]$$

with $g \in \{m, w\}$ men or women, t event time, s year, and $\tilde{Y}$ the prediction from age and year alone. P_t is the **child penalty**: the gap that opens, as a share of women's counterfactual earnings.

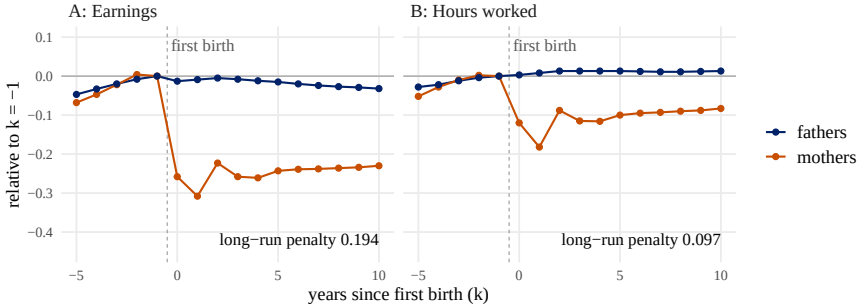
Kleven, Landais, and Søgaard: Reading the Child Penalty

$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w)/E[\tilde{Y}_{ist}^w \mid t]$, step by step, with round numbers (hypothetical, in DKK):

1. Fathers, five years after the first birth: earnings at their $t = -1$ level, so $\hat{\alpha}_5^m = 0$.
2. Mothers: 40,000 below their $t = -1$ level, so $\hat{\alpha}_5^w = -40{,}000$.
3. Gap: $\hat{\alpha}_5^m - \hat{\alpha}_5^w = 0 - (-40{,}000) = 40{,}000$.
4. Scale by what mothers would have earned without children (age and year alone predict 200,000): $P_5 = 40{,}000/200{,}000 = 0.20$.

$P_5 = 0.20$: five years on, children have opened a gap of 20% of women's counterfactual earnings.

Kleven, Landais, and Søgaard: Results



Their Figure 1 A–B, values read off the figure (±0.01) (figs/w10_kleven.R).

- Mothers' earnings fall almost 30% at birth and settle about 20% below the counterfactual; fathers' do not move. Long-run penalty: earnings 0.194, hours 0.097, participation 0.130, wage rate 0.091.
- The share of gender earnings inequality due to children rose from about 40% (1980) to about 80% (2013).

Event Studies: Specification Details That Matter

Three choices inside the specification change the picture; report each.

- **Normalization**: one lead must be omitted (here $k = -1$). Every τ_k is relative to it, so the picture shifts if you change the reference period. Say which you used.
- **Binning the endpoints**: with unbalanced exposure, far leads and lags are estimated off few units. Bin them ($k \leq -5$, $k \geq 5$) or restrict to a balanced window, and say which.
- **Never-treated units** anchor the time effects λ_t.

The Event Study on the Staggered Toy ★

The staggered toy: early starts at $t = 2$ with effects **1, 2, 3** (growing), late at $t = 4$ with effect 1. The forbidden 2×2, late vs. early, $t = 3 \to 4$:

	change, $t = 3 \to 4$	static TWFE reads it as	event study reads it as
late (treated)	$0 \to 1$: $+1$	β	τ_0
early (control)	$2 \to 3$: $+1$	$\beta - \beta = 0$	$\tau_2 - \tau_1$
2×2	$1 - 1 = 0$	$0 = \beta$	$0 = \tau_0 - (\tau_2 - \tau_1)$

- **Static TWFE** thinks early's effect is constant, so early's $+1$ must be the common trend. It reads the 2×2 as $\beta = 0$, and is pulled down: 1.10, not 1.75.
- **The event study** knows early moved from $k = 1$ to $k = 2$, and that its effect grew by $\tau_2 - \tau_1 = 3 - 2 = 1$. It subtracts that growth: $\tau_0 = 0 + 1 = 1$. Correct.
- The TWFE event study gives exactly 1, 2, 3: it fixes the growing-effect problem.

What the Event Study Still Assumes ★

To subtract early's growth from late's comparison, the event study assumes every cohort follows the same path τ_0, τ_1, τ_2: one τ_k for all.

Break it: keep early at 1, 2, 3, but late gains 3 at $k = 0$. Early vs. never says $\tau_0 = 1$, late vs. never says 3; one τ_0 cannot be both, and the misfit leaks:

k	-3	-2	0	1	2
truth (average over cohorts)	0	0	2	2	3
TWFE event study	-0.75	-0.25	2.00	2.75	3.25
clean blocks (vs. never, base $g-1$)	0	0	2	2	3

- A **fake pre-trend** and wrong post effects, from treatment effects alone: the forbidden 2×2 is still in the regression (Sun and Abraham 2021; figs/w10_toy_staggered.R).
- Why the leads move: late jumps 3, the shared $\hat{\tau}_0$ is 2, so OLS raises late's level ($\hat{\alpha} = 0.75$); the leads go negative to pull its before-years back to 0.
- Fix: give each cohort its own path, then average by k (next).

Sun–Abraham: the Same Blocks, as One Regression

Sun and Abraham (2021)'s fix for the event study: do not force one τ_k on all cohorts. Interact event time with cohort:

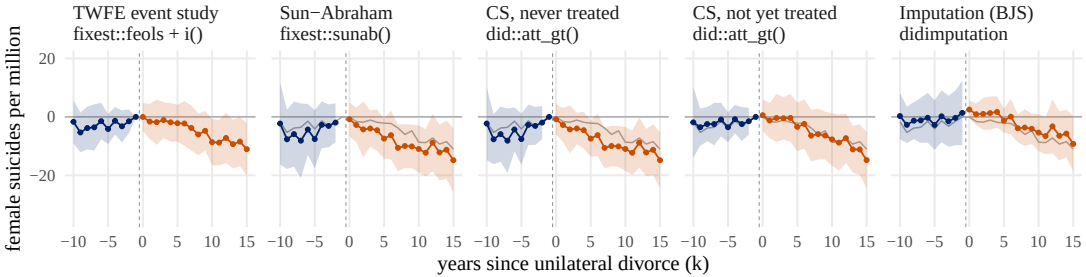
$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_{k \neq -1} \delta_{gk} \mathbb{1}\{G_i = g\} \mathbb{1}\{t - g = k\} + \epsilon_{it}$$

- δ_{gk}: cohort g's effect k years after starting, one coefficient per block.
- Then average across cohorts at each k, weighted by cohort size (`fixest::sunab`).

	Sun–Abraham	Callaway–Sant'Anna
how	one regression	separate 2×2s
controls	never (or last) treated	never *or* not yet treated
covariates	linear, in the regression	OR, IPW, or DR in each block

Same controls, no covariates: identical estimates (divorce data: both -6.5 over $k = 0$–10; one overall number, both -10.56).

The Event Studies, Side by Side



- Sun–Abraham and Callaway–Sant'Anna (never-treated controls) give the *same* path: -6.5 on average over $k = 0$–10, -14.9 at $k = 15$. TWFE: -3.1.
- CS with not-yet-treated controls: -3.1 on average. Later adopters are controls early on; by $k = 15$ only never-treated states remain, and it meets CS never (-14.8).
- Imputation uses every untreated year as baseline: -1.0 on average. Leads are not flat (CS never, $k = -5$: -7.6), so the choice of baseline years matters (figs/w10_divorce_es.R).

Choices Every Staggered DiD Makes

choice	options	consequence
comparison group	never treated; not yet treated	not-yet uses more data but needs no anticipation and parallel trends with future adopters
baseline	$g-1$ only; all pre-periods	all pre-periods is efficient if trends are parallel throughout, biased if not
aggregation	simple; by event time; by cohort; by calendar time	each is a different estimand; say which
anticipation	shift g back by δ periods	otherwise effects leak into the baseline
covariates	none; DR within each $ATT(g,t)$	use pre-treatment values only
inference	cluster; bootstrap; randomization	few treated cohorts need care

Efficiency when adoption timing is as good as random: Roth and Sant'Anna (2023b) (`staggered`). A design-based reading of TWFE under random timing: Athey and Imbens (2022).

Beyond 0/1

TWFE Gets Run on Any Treatment ★

In sociology $Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it}$ is run on almost anything that changes within a person:

- a **count or dose**: number of children (Budig and England), hours worked, neighborhood poverty rate;
- a status that **switches on and off**: married, cohabiting, union member, employed.

How $\hat{\beta}$ is read: "one more unit of D raises Y by β." That reading needs four things:

1. the effect is **linear** in D: the first child does what the third one does;
2. it is the **same for every person**: no one chose D because they gain more;
3. it is the **same at every time** since the change: no growth or fade-out;
4. **no carryover**: once D goes back down, the past dose stops mattering.

What $\hat{\beta}$ actually is: a weighted average of comparisons between people whose D changed by different amounts at different times, already-changed people included as controls.

Doses: a Continuous Treatment

Callaway, Goodman-Bacon, and Sant'Anna (2025). Units receive dose d (a minimum-wage increase of \$1 or \$2); some get $d = 0$.
$\text{ATT}(a \mid d) = E[Y(a) - Y(0) \mid D = d]$: the effect of dose a on the units that got dose d. Here $a = d$; $a \neq d$ (e.g. $\text{ATT}(1 \mid 2)$) is never observed (next slides).

Hypothetical: three groups of states, employment change after the law.

group	change	DiD against $d = 0$
$d = 0$	-1.0	
$d = \$1$	-0.5	$\text{ATT}(1 \mid 1) = 0.5$
$d = \$2$	0.4	$\text{ATT}(2 \mid 2) = 1.4$

- Each row is an ordinary DiD: ordinary parallel trends (each dose group vs. $d = 0$) identifies 0.5 and 1.4.
- The tempting next step: "the second dollar adds $1.4 - 0.5 = 0.9$." That compares *different states*.

Doses: What TWFE Reports

Two periods, before and after the law; $D_{it} = d_i$ after, 0 before. With two periods, TWFE is the slope of the change (after − before) on the dose. Same three states (figs/w10_twfe_many.R):

2×2 (before → after)	difference in changes	dollars apart	per dollar
\$1 vs. \$0	$-0.5 - (-1.0) = 0.5$	1	0.5
\$2 vs. \$1	$0.4 - (-0.5) = 0.9$	1	0.9
\$2 vs. \$0	$0.4 - (-1.0) = 1.4$	2	0.7
TWFE $\hat{\beta}$	slope through the three points: $1.4/2 = 0.5(0.5) + 0.5(0.9) = 0.7$		

- "Each dollar raises employment by 0.7." No dollar does: the first adds 0.5, the step to \$2 adds 0.9; the \$2 vs. \$0 2×2 is the two steps added.
- OLS sets the weights on the steps (here 0.5 each, from how many states sit at each dose). And the 0.9 compares *different states* (next).

Doses: Why Comparing Doses Needs More

Split the difference between the two dose groups:

$$\underbrace{\text{ATT}(2 \mid 2) - \text{ATT}(1 \mid 1)}_{0.9} = \underbrace{\text{ATT}(2 \mid 2) - \text{ATT}(1 \mid 2)}_{\text{causal: a 2nd dollar, same states}} + \underbrace{\text{ATT}(1 \mid 2) - \text{ATT}(1 \mid 1)}_{\text{selection: different states, same dose}}$$

- $\text{ATT}(1 \mid 2)$, what \$1 would have done in the \$2 states, is never observed. If those states respond more (say 0.8), the 0.9 is 0.6 causal +0.3 selection.
- **Strong parallel trends** assumes the selection term is zero: every dose group would respond to a dose as any other would. It rules out selection on gains.
- TWFE's 0.7 averages this mixed 0.9 with the 0.5. `contdid` estimates $\text{ATT}(d \mid d)$ dose by dose and averages them with weights you choose.

Doses With Staggered Timing: What to Do Instead

Callaway, Goodman-Bacon, and Sant'Anna (2025): units start at different dates *and* with different doses. Same logic as Callaway–Sant'Anna, with a dose added:

1. **Blocks**: $ATT(d \mid g, t)$, units of cohort g that got dose d, against units not yet treated at t, from $g-1$ to t. A clean 2×2 for each dose.
2. **Aggregate** across cohorts and periods: a dose-response curve $ATT(d \mid d)$, the effect of each dose on the units that got it.
3. **Compare doses** ("what does one more unit do?") only under strong parallel trends (previous slide). Report which you assume.

- R: `contdid`. Treatments that change many times, or go up and down (children, union): de Chaisemartin and D'Haultfœuille (2026), `DIDmultiplegtDYN`.

Treatments That Switch On and Off

Union membership, unemployment, living with a partner: units enter and leave treatment, and past treatment may still matter. Three workers (1 = union member):

worker	2001	2002	2003	2004	2005
A	0	0	**1**	1	**0**
B	0	0	0	0	0
C	1	1	1	**0**	0

- **Clean**: A joins in 2003; B had the same history (0, 0) and does not change. A minus B, 2002 to 2003, is a 2×2 for joining.
- **Not clean**: TWFE also uses C (who left in 2004) and A after it leaves in 2005, assuming no carryover: having been a member changes nothing once you leave.
- de Chaisemartin and D'Haultfœuille (2026): compare only units with the same history that have not yet changed; DID_ℓ is the effect after ℓ periods (`DIDmultiplegtDYN`). Cost: fewer comparisons, wider intervals.

Looking Ahead

- Fixed effects difference away time-invariant confounders, bought with precision and with strict exogeneity; DiD adds parallel trends.
- The staggered-adoption literature is a case study in how a method can be standard for decades and still be wrong. Read papers published before 2021 with this in mind, including your own field's classics.
- **Next week**: what to do when there is only *one* treated unit, and "the control group" has to be constructed.
- Notice that conditional parallel trends brought Week 6 back inside a DID: once trends are only parallel given covariates, you need a weighting or outcome model again, and all of Week 6's cautions return with it.

Problem Set 4

Assigned today, due Monday November 9. Covers Weeks 7 and 9.

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